Tuberculosis Diagnostic Syste ×

localhost:5173/patient/MED-2026-DEMO-0002/features

TB **TB-DOTS CDSS**
CAR Region

- Patient Overview
- Dashboard
- New Patient

BAGUIO CITY 1%

Increases failure risk Decreases failure risk

🧠 **AI Clinical Interpretation** MedGemma

1. Risk Assessment Summary

The 83% treatment failure probability indicates a high risk for this patient to experience treatment failure during their TB treatment course.

2. Key Risk Factors

- **Microscopy Result (patient value: 2+)**: This finding had a +16.0% impact, increasing the failure risk. Because this patient has a 2+ result, it suggests a higher bacterial load, which is associated with poorer treatment outcomes and increased transmission potential.
- **Registration Group (patient value: RELAPSE)**: This factor contributed positively (+15.0%) to the failure probability. As a relapse case, the patient likely had previous TB exposure or incomplete treatment, indicating underlying vulnerabilities that may not have been fully addressed by prior therapy, increasing the risk of recurrence and treatment failure.
- **Days to Treatment**: This factor increased the failure risk by +14.0%. Starting treatment 12 days after diagnosis suggests a delay in initiating care, potentially leading to poorer initial bacterial load or progression of disease before effective treatment began, thus raising the probability of failure.
- **Type (patient value: DRTB)**: The patient's DRTB status had a +13.0% impact on the failure risk. Drug-resistant TB inherently poses a greater challenge for treatment success compared to drug-susceptible TB due to the need for more complex and potentially longer regimens, increasing the likelihood of treatment failure if not managed appropriately.
- **Bacteriologic Status (patient value: Bacteriologically-confirmed TB)**: This factor contributed positively (+8.0%) to the failure risk. Being bacteriologically confirmed means the patient has detectable TB bacteria, which is a prerequisite for effective treatment but also signifies active disease that may be more difficult to control, increasing the risk of failure compared to patients who are not yet confirmed.

3. Why the Model Reached This Conclusion

The model reached this high probability score because the patient presents with multiple significant risk factors combined. The presence of DRTB (Drug-Resistant TB) is a major contributor due to the inherent difficulty in treating resistant strains. The patient's microscopy result of 2+ indicates a substantial bacterial load, further complicating treatment. Furthermore, being a relapse case suggests potential issues from previous attempts or underlying vulnerabilities. The delay in starting treatment (12 days) likely allowed the disease to progress somewhat before therapy began. The combination of these factors, particularly DRTB and the high bacterial load indicated by the microscopy result, significantly elevates the probability that this patient will not achieve a cure, resulting in a 83% failure prediction.

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